

MLFF CUEQ-PHASE1 Development-Host Qualification

mdstats 0.20.189a0 / architecture revision 56

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Result

Control-plane gate: PASS

Positive GPU qualification: DEFERRED to FINAL-GPU1

Phase-separated CuEq training authorization: NOT YET AUTHORIZED

CUEQ-PHASE1 is implemented as a training-only accelerator qualification. Source-foundation inference, DATA6, pseudolabel generation, and source evaluation remain on original MH-1/omat_pbe/e3nn. Only TRAIN2 may vary between e3nn and cueq_pure in the final paired campaign.

Development-host evidence

The retained CUEQ-DEP1 development-host record is negative by design: `passed=false`, CUDA device count 0, and blockers `cueq-core_import`, `cueq-torch_import`, `cueq-ops_import`, `torch_cuda_available`, `cuda_device_inventory`. Its content digest is 45a3d9aa919e557388dc6858feb64b7009f4eeaa838c097d4456f7f29b56f069.

The phase-1 reducer is also fail-closed: `passed=false`, with blockers `CUEQ_DEP1_RUNTIME_FREEZE`, `short_paired_adaptation_representative_full_pair_missing`. A short-only result cannot authorize training, and no short or representative full CuEq trajectory was executed on this development host.

Foundation identities

Foundation	Required head	SHA-256
MACE-MH-1	omat_pbe	ec00a2705854622fbbd898ccfb7701072fcd67470
MACE-MPA-0-medium	default	75428afe3a1d7d8062e19bcaabd5c433623cabf30

Both uploaded model identities pass the release lock. The FINAL-GPU1 preflight schema is `mdstats.mlff-final-gpu1.preflight` and remains `deferred_not_executed` on this CPU-only host.

CPU/control-plane validation

The focused adjacent suite passes **26/26 tests** with the real supplied foundation-model paths enabled. Serialization/tamper guards, protocol mismatch rejection, hard-decision mismatch rejection, short-only non-authorization, positive synthetic reducer behavior, FINAL-GPU1 handoff, and revision/schema synchronization all pass. `compileall` and stale current-version/schema scans also pass.

Final-GPU1 obligations

A release-authorizing CUEQ-PHASE1 record still requires, on the same positive CUEQ-DEP1 runtime:

1. one passing 5-10 epoch paired e3nn versus pure-CuEq adaptation (default 8 epochs); and
2. at least one passing representative full paired trajectory under identical frozen scientific/training inputs.

Final checkpoint bytes are not required to match. Existing replay-retention, finiteness, checkpoint-admissibility, selected-head extraction, EVAL2, and available physical-verification decisions must agree and pass. Performance/VRAM telemetry is diagnostic and cannot relax scientific thresholds.

Authorization boundary and next gate

This release does **not** authorize CuEq training yet, CuEq source execution, CuEq DATA6/pseudolabel generation, or a generated-default change. The next implementation gate is optional **CUEQ-PHASE2**, selected-head CuEq source-execution/DATA6 qualification, while the actual GPU evidence for **CUEQ-PHASE1** remains consolidated into **FINAL-GPU1**.